

Physics — Unit 6: Work and Energy

Seemab | Class 9 | FBISE Board Exam Style | Attempt: 2026-06-22
Exam: physics/exams/exam-04-ch6 | Marked: 2026-06-22

55.5/65

TOTAL SCORE

85.4%

PERCENTAGE

A

GRADE

12/12

SECTION A — MCQS

29/33

SECTION B — SHORT

14.5/20

SECTION C — LONG

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Topic	Correct	Seemab	Result
1	SI unit of work	C — Joule	C	✓
2	Work done pushing immovable wall	C — Zero	C	✓
3	PE of 5N ball lifted 5m	B — 25 J	B	✓
4	KE depends on mass and square of	B — Speed	B	✓
5	KE of 4kg at 5 m/s	B — 50 J	B	✓
6	Law of Conservation of Energy — can only	B — Change forms	B	✓
7	SI unit of power	C — Watt	C	✓
8	Power when 100J done in 5s	B — 20 W	B	✓
9	1 horsepower equals	D — 746 W	D	✓
10	Renewable energy resource	C — Sunlight	C	✓
11	Engine efficiency (400J in, 120J out)	C — 30%	C	✓
12	Practical engine efficiency never equal to	D — 1 (100%)	D	✓

Section A: 12/12 — Perfect score. Physics MCQ streak continues (4th consecutive perfect Section A across all Physics exams).

SECTION B — Short Answer Questions (11 × 3 = 33 Marks)

Q.2(i) OR — Muhammad Taha pushes box 150N through 5m. Find work done.

3 / 3

Data correctly identified: $F=150\text{N}$, $d=5\text{m}$. Formula $W=F \times d$ applied. Answer: $150\text{N} \times 5\text{m} = 750\text{J}$. Units stated as Nm/J. Full marks.

Q.2(ii) OR — Babar Azam: ball 120g, KE=50J. Find speed.

3 / 3

Mass conversion: $120\text{g} \div 1000 = 0.12\text{ kg}$ shown explicitly. Formula $E_k = \frac{1}{2}mv^2$ rearranged correctly: $v^2 = 2E_k/m = 100/0.12 = 833$. $v = \sqrt{833} = 28.7\text{ m/s}$. (Key: 28.9 m/s using 833.3; difference is rounding only — no deduction.) All steps present, method fully correct.

Q.2(iii) OR — Kamil, 60kg, jumps 1.5m. Find gravitational PE at top.

3 / 3

$E_p = mgh = 60 \times 9.8 \times 1.5 = 882\text{J}$. All three variables correctly substituted. Answer correct.

Q.2(iv) OR — Football 0.45kg kicked at 22m/s. Find max height (no friction).

3 / 3

Conservation of energy: $KE = PE \rightarrow \frac{1}{2}mv^2 = mgh \rightarrow v^2/(2g) = h$. Masses cancel correctly; $h = (22)^2/(9.8 \times 2) = 484/19.6 = 24.7\text{m}$. Answer and method both correct.

Q.2(v) Main — Energy conversion: (a) dam and (b) cyclist going uphill.

3 / 3

Definition of energy conversion given. (a) Dam: gravitational PE of water at height converts to KE, falls on turbine, produces electricity — chain clearly stated. (b) Cyclist: stored chemical energy allows work against gravity; at top possesses gravitational PE; going downhill KE increases without pedalling — full conceptual chain. Both examples complete.

Q.2(vi) Main — Differentiate renewable and non-renewable resources with two examples each.**1 / 3**

Definitions swapped: under the "Renewable" column, Seemab wrote "Renewable resources is the resource that cannot be replaced naturally in a short time" — this is the non-renewable definition. The definitions in both columns are exchanged. Each definition independently loses its mark. -2 marks.

Environmental benefit of renewables not stated (no sentence about greenhouse gases / sustainability). No additional mark lost since only 3 marks total and examples earn the 1 remaining.

Examples are correct: Renewable — sunlight, flowing water; Non-renewable — coal, oil/natural gas. 1 mark awarded for examples.

Drill note: Renewable = replenished naturally and rapidly (sun shines again, water flows again). Non-renewable = formed over millions of years; finite; cannot be replaced in a human timescale. The key distinguishing word is the RATE of replacement, not whether replacement happens at all.

Q.2(vii) Main — Fossil fuels: formation, three types, combustion equation for CH₄.**3 / 3**

Formation: remains of million-years-old plant life (coal) and aquatic animal life (gasoline/oil); hydrocarbon molecules of hydrogen and carbon — correct. Three types: Coal, Natural gas, Oil — all correct. Combustion equation: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{energy}$ — balanced correctly. Full marks.

Q.2(viii) OR — Solar radiation as energy source: two methods, one advantage, one disadvantage.**1 / 3**

Definition incorrect: wrote "Energy that directly converts sunlight into electricity is known as solar radiation." Solar radiation is sunlight/electromagnetic energy from the Sun — the conversion process described is what solar cells do, not what solar radiation is. -1 mark for definition.

Advantage and disadvantage not written. The answer ends after listing the two methods. -1 mark.

Two methods correctly identified: (1) Solar photovoltaic cells, (2) Thermal solar system. 1 mark awarded.

Correct definition: Solar radiation is energy emitted by the Sun in the form of electromagnetic waves (light and heat) that reaches Earth. Advantage: clean, renewable, no greenhouse gases. Disadvantage: intermittent (depends on weather and time of day).

Q.2(ix) OR — Tesla engine: $W=3.69 \times 10^7 \text{ J}$ in 1 min. Find power in W and hp.**3 / 3**

$t=1 \text{ min}=60 \text{ s}$ stated. $P=W/t=(3.69 \times 10^7)/60=6.15 \times 10^5 \text{ W}$. Convert: $P=6.15 \times 10^5/746=825 \text{ hp}$. All steps clearly shown. Full marks.

Q.2(x) OR — 2hp motor: $E_{\text{in}}=2238 \text{ J}$, lifts 10kg to 10m. Find output energy and efficiency.**3 / 3**

$E_{\text{out}}=mgh=10 \times 9.8 \times 10=980 \text{ J}$. Efficiency: $\eta=(980/2238) \times 100=43.8\%$. Both values correct. Note: data line shows "22.38J" (decimal slip in copying) but all calculations use the correct 2238J value throughout — no penalty as the working is unambiguous and correct.

Q.2(xi) OR — 5kg steel ball dropped from 12m. Find KE at ground and speed.**3 / 3**

$E_p=mgh=5 \times 9.8 \times 12=588 \text{ J}$. By conservation: $E_k=588 \text{ J}$. Speed: $\frac{1}{2}(5)v^2=588 \rightarrow v^2=235.2 \rightarrow v=15.33 \text{ m/s}$ (key: 15.34 m/s; rounding difference, no deduction). All working shown correctly.

Section B Total: 29/33 | Deductions: Q.2(vi) definitions swapped -2, Q.2(viii) definition wrong -1 + adv/disadv missing -1**SECTION C — Long Answer Questions (4 × 5 = 20 Marks)****Q.3(i) Main — Work definition, SI unit, two zero-work conditions, derive $E_k=\frac{1}{2}mv^2$, work-energy theorem, 2.0kg rock numerical.****4.5 / 5**

Definition: "Work is said to be done when a force displaces a body in its own direction." SI unit: Joule (J), $1\text{J}=1\text{N}\cdot\text{m}$ — correct, 1 mark.

Zero-work condition (1): Seemab wrote "Parallel (force and displacement), $\theta=0^\circ$, $W=FS\cos 0^\circ=FS$ " — this gives MAXIMUM work ($W=Fd$), not zero work. The first condition for zero work is either (a) displacement is zero or (b) force perpendicular to displacement. Only condition (2) — perpendicular, $\theta=90^\circ$, $W=FS\cos 90^\circ=0$ — is correct. -0.5 mark for one wrong condition.

Derivation of $E_k=\frac{1}{2}mv^2$: uses work-done= $Fd=(ma)(v_{\text{avg}} \times t)=m(v/t) \times (v/2)t=\frac{1}{2}mv^2$ — the textbook graphical method steps are correctly followed. 1 mark.

Work-Energy Theorem: $W=\Delta E_k$ — stated correctly with equation. 1 mark.

Numerical: height fallen=15m, remaining=20-15=5m. $KE=2 \times 9.8 \times 15=294 \text{ J}$. Remaining $PE=2 \times 9.8 \times 5=98 \text{ J}$. Verification: $294+98=392 \text{ J}=mgh_{\text{total}}=2 \times 9.8 \times 20=392 \text{ J}$ (shown explicitly). 1 mark.

The only slip: parallel force/displacement (0°) gives maximum work, not zero. The two correct zero-work conditions are: (1) displacement is zero — pushing a wall that does not move; (2) force is perpendicular to displacement — centripetal force on a satellite in orbit.

Q.3(ii) OR — Energy conversion and dissipation: three examples, where 80% goes in petrol car, why electric cars more efficient.

3.5 / 5

Definition: energy conversion = changing energy from one form to another. 1 mark.

Example 1 — Dam (hydroelectric): gravitational PE of water at height converts to KE then to electricity via turbine. Conversion chain present.

Example 2 — Cyclist going uphill: stored chemical energy allows work against gravity; at top possesses gravitational PE; going downhill KE increases without pedalling. Both examples complete and accurate. 1.5 marks for 2 full + 1 partial example.

Third example (petrol car engine conversion chain) is discussed as a dissipation context but the energy conversion chain (chemical PE → KE + heat) is not stated explicitly as a textbook conversion example. -0.5 marks (partial credit).

Where 80% goes: heat from engine, exhaust gases, friction in mechanical parts — "dissipated energy" cited. 1 mark.

Electric cars more efficient: not addressed anywhere in the answer. -1 mark.

Missing point: electric motors convert electrical energy directly to mechanical motion with far less heat loss compared to internal combustion engines, hence much higher efficiency (80%+ vs 17-21%).

Q.3(iii) Main — Define power, $P=W/t$, prove $1\text{kWh}=3.6\times 10^6\text{J}$, find mass M , convert to hp.

3.5 / 5

Power definition: "the time rate at which work is done or energy is consumed." Formula $P=W/t$ (also $P=E/t$). SI unit: Watt (W), $1\text{W}=1\text{Js}^{-1}$, named after James Watt. All three elements (definition + formula + unit) present. 1 mark.

kWh defined as "unit of energy used in electricity billing; energy converted or consumed at the rate of 1kW for 1 hour." Definition correct. 0.5 mark.

Proof that $1\text{kWh}=3.6\times 10^6\text{J}$ not shown. Required: $1\text{kWh}=1000\text{W}\times 3600\text{s}=3,600,000\text{J}=3.6\times 10^6\text{J}$. Seemab only states the definition, skips the arithmetic proof. -0.5 mark.

Numerical — Find mass M : $P=1960\text{W}$, $h=2\text{m}$, $t=3\text{s}$. Work done $=P\times t=1960\times 3=5880\text{J}$. Then $W=mgh \rightarrow m=W/(gh)=5880/(9.8\times 2)=5880/19.6=300\text{kg}$. Correct. 1 mark.

Convert 1960W to hp: $1960/746=2.62\text{hp}$. Correct. 1 mark.

Proof of $1\text{kWh}=3.6\times 10^6\text{J}$ is a single calculation: $E=P\times t=1000\text{W}\times 3600\text{s}=3,600,000\text{J}=3.6\times 10^6\text{J}$. Always write this when "prove" is in the question.

Q.3(iv) Main — Renewable vs non-renewable table, fossil fuels (formation/types/extraction/combustion), two environmental problems, why shift to renewables.

3 / 5

Definitions swapped again (same error as Q.2(vi)): under "Renewable resources" Seemab wrote "is the resource that cannot be replaced naturally in a short time" — the non-renewable definition. Both column definitions incorrect. -1 mark for definitions.

Examples: Renewable — sunlight, solar radiation, wind, flowing water. Non-renewable — coal, fossil fuels, oil, natural gas. Correct examples in both columns. 1 mark.

Fossil fuels — Formation: remains of million-years-old plant life (coal) and aquatic life (gasoline, oil, natural gas); hydrocarbon molecules of hydrogen and carbon. Types: Coal, Oil, Natural gas. Combustion equation: $\text{CH}_4+2\text{O}_2\rightarrow\text{CO}_2+2\text{H}_2\text{O}+\text{energy}$ — balanced correctly. Formation + types + equation all present. 2 marks.

Extraction methods: only "coal is mined" mentioned — very thin. -0.5 from partial credit on fossil fuels subsection (absorbed into the 2 mark allocation above as partial).

Two environmental problems not stated. Why society must shift to renewables not addressed. Answer ends after the combustion equation. -1 mark.

Missing: environmental problems — (1) burning fossil fuels releases $\text{CO}_2/\text{SO}_2/\text{NO}_2$ causing global warming, acid rain; (2) oil spills and mining damage ecosystems. Why shift: fossil fuels will run out; renewables are sustainable and cleaner. Always write these closing points before stopping.

Section C Total: 14.5/20 | Deductions: Q.3(i) wrong zero-work condition -0.5; Q.3(ii) third example partial -0.5 + electric cars missing -1; Q.3(iii) kWh proof missing -0.5; Q.3(iv) definitions swapped -1 + environmental problems/shift missing -1

Strengths and Areas for Improvement

STRENGTHS THIS ATTEMPT

- + Perfect MCQs (12/12) — all conceptual traps (zero work, conservation, horsepower, efficiency limit) navigated correctly.
- + Numerical fluency is excellent: all 6 calculation-based Section B OR questions correct; mass-conversion (g→kg), energy conservation, $P=W/t$, efficiency formula all applied without error.
- + Work-energy theorem stated precisely ($W=\Delta E_k$) and numerical verification shown explicitly ($294+98=392\text{J}$) checked against mgh_{total} .
- + Fossil fuels topic complete: formation, three types, and balanced combustion equation all present in both Q.2(vii) and Q.3(iv).
- + Energy conversion examples (dam and cyclist) fully described with complete conversion chains in three separate questions (Q.2(v), Q.3(ii) examples, Q.3(iv) context).
- + Tesla engine power calculation: scientific notation maintained correctly through division ($3.69\times 10^7\div 60=6.15\times 10^5$); hp conversion correct.

RECURRING ERRORS TO FIX

- ! **Renewable definition swapped (new recurring, 2 questions this paper):** Q.2(vi) and Q.3(iv) both show the renewable definition placed in the non-renewable column and vice versa. Key: Renewable = replaced rapidly by natural processes. Non-renewable = formed over millions of years, cannot be replaced in human timescale.
- ! **Zero-work conditions (Q.3(i)):** "Parallel force and displacement" gives MAXIMUM work, not zero. The two zero-work conditions are (1) displacement=0 and (2) force perpendicular to displacement.
- ! **Solar radiation definition (Q.2(viii)):** Solar radiation is electromagnetic energy from the Sun, not the process of converting it. The conversion is what solar cells do.
- ! **kWh proof omitted (Q.3(iii)):** When "prove" appears in the question, the arithmetic must be shown:
 $E=1000\text{W}\times 3600\text{s}=3.6\times 10^6\text{J}$.
- ! **Rushed endings on long questions (Q.3(ii) and Q.3(iv)):** Both long answers stop before reaching all required sub-parts. Q.3(ii) missing electric cars comparison; Q.3(iv) missing environmental problems and reasons to shift. This is an old pattern returning — check all bullet points in the question before putting pen down.
- ! **Solar radiation adv/disadv (Q.2(viii)):** Advantage and disadvantage completely missing despite the question explicitly

requiring one of each.

Physics Exam Comparison (This Subject Only)

#	Date	Exam	MCQ	Sec B	Sec C	Total	%	Grade
1	7 Apr 2026	exam-01-ch1-2	12/12	32.5/33	19/20	63.5/65	97.7%	A+
2	10 Apr 2026	exam-02-ch3-4-9	12/12	31/33	19/20	62/65	95.4%	A+
3	18 May 2026	exam-03-ch5	12/12	33/33	20/20	65/65	100%	A+
4	22 Jun 2026	exam-04-ch6	12/12	29/33	14.5/20	55.5/65	85.4%	A

Physics subject average (4 exams): $(63.5+62+65+55.5)/260 = 246/260 = 94.6\%$. MCQ record: 48/48 (100%) — all four Physics exams perfect in MCQs.